WEEK 1

Mobile Application Development

Platforms • Ecosystems • Flutter • Setup

Session 1: Overview & Platforms

Session 2: Flutter Intro

Session 3: SDK Setup

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Overview of Mobile Application Development & Mobile Platforms

CONTINUE

Learning Objectives

01	Understand what mobile applications are and how they work in daily life
02	Know the major mobile platforms — Android and iOS
03	Understand the full mobile app ecosystem and its components
04	Recognize real-world examples of mobile apps across different categories

What Are Mobile Applications?

A mobile application is software program designed to run on smartphones and tablets
Mobile apps solve real-life problems and make daily task easier

Social Media	E-Commerce	Educational	Gaming
WhatsApp Instagram	Amazon Daraz	Duolingo Corsera	PUBG Subway Surfers

What Is Mobile Application Development?

The process of creating software application for mobile such as smartphone and tablets
Mobile apps solve real-life problems and make daily task easier

Lifecycle:

1. Idea/Problem identification
2. Planning
3. Designing
4. Development
5. Testing
6. Deployment
7. Maintenance

Major Mobile Platforms

A mobile platform is the operating system that runs mobile apps
Just like computer have Windows or macOS

Android

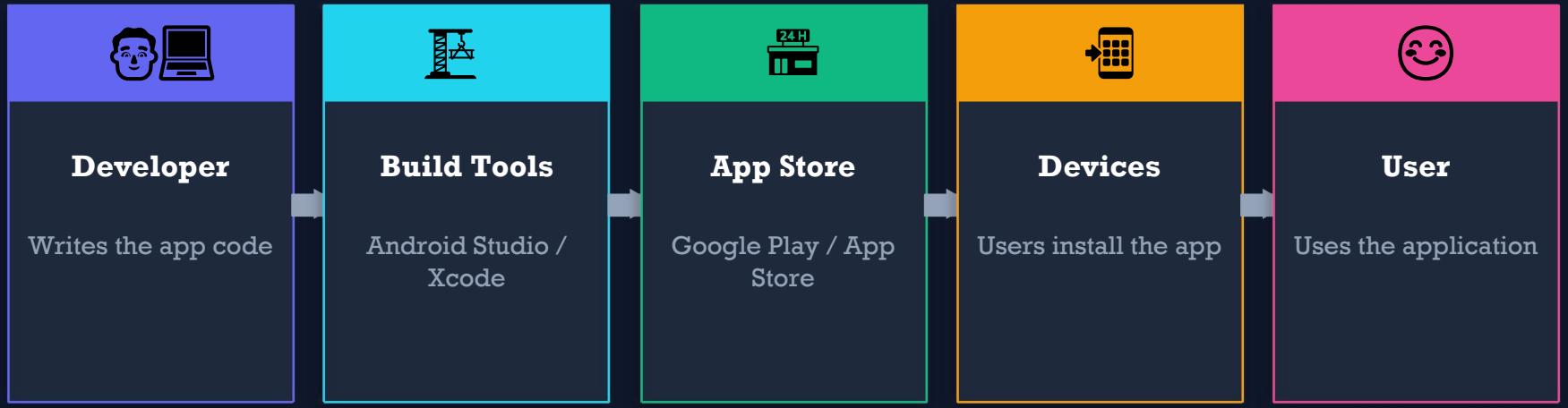
Developer:	Google
Ecosystem:	Open Source
Devices:	Samsung, Xiaomi, Oppo, Vivo...
Language:	Kotlin / Java
Market Share:	~72% Globally

VS

iOS

Developer:	Apple
Ecosystem:	Closed / Controlled
Devices:	iPhone, iPad only
Language:	Swift / Objective-C
Market Share:	~28% Globally

Mobile App Ecosystem



5 Core Components: Mobile OS | App Store | Dev Tools | Developers | Users

Key Insight: The ecosystem is a complete chain — from the developer's code editor to the user's fingertip. Every link matters for app success.

Types of Mobile Applications

⚡ Native Apps

Built specifically for one platform. Offers the best performance and full device access.

✓ Pros:

Best performance
Full hardware access
Smooth UI

⚠ Cons:

Separate codebase needed
Higher cost

Android → Kotlin/Java
iOS → Swift/Obj-C

🌐 Web Apps

Run inside a mobile browser. No installation needed — accessible via URL.

✓ Pros:

No install required
Cross-platform
Easy updates

⚠ Cons:

Limited device access
Needs internet

HTML • CSS • JavaScript

🔀 Cross-Platform Apps

One codebase, multiple platforms. Balance of speed and reach.

✓ Pros:

Single codebase
Faster dev
Lower cost

⚠ Cons:

Slight performance gap
Framework dependency

Flutter • React Native
Xamarin

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Native vs Cross-Platform Development & Flutter Introduction

CONTINUE

Native Development

📌 Native apps are built specifically for ONE platform using the platform's official language and tools

Android Native

Primary Language:	Kotlin
Legacy Language:	Java
IDE:	Android Studio
Build Tool:	Gradle
UI Toolkit:	Jetpack Compose / XML

iOS Native

Primary Language:	Swift
Legacy Language:	Objective-C
IDE:	Xcode
Build Tool:	Xcode Build
UI Toolkit:	SwiftUI / UIKit

Cross-Platform Development

Popular Cross-Platform Frameworks:

Flutter

Language: Dart
By: Google
Rating: ★★★★★

React Native

Language: JavaScript
By: Meta
Rating: ★★★★☆

Xamarin

Language: C#
By: Microsoft
Rating: ★★★☆☆

⚖ *Tradeoff: Cross-platform = faster & cheaper, but native = best performance & device integration*

Introduction to Flutter

Flutter

by Google

Language: Dart

Targets:

- Android
- iOS
- Web
- Desktop

Hot Reload

See changes instantly without restarting the entire app. Saves massive development time.

Widget-Based

Everything in Flutter is a Widget. Text, Buttons, Images, Layouts — all are widgets.

Beautiful UI

Pixel-perfect, expressive UI with Material Design and Cupertino (iOS-style) components.

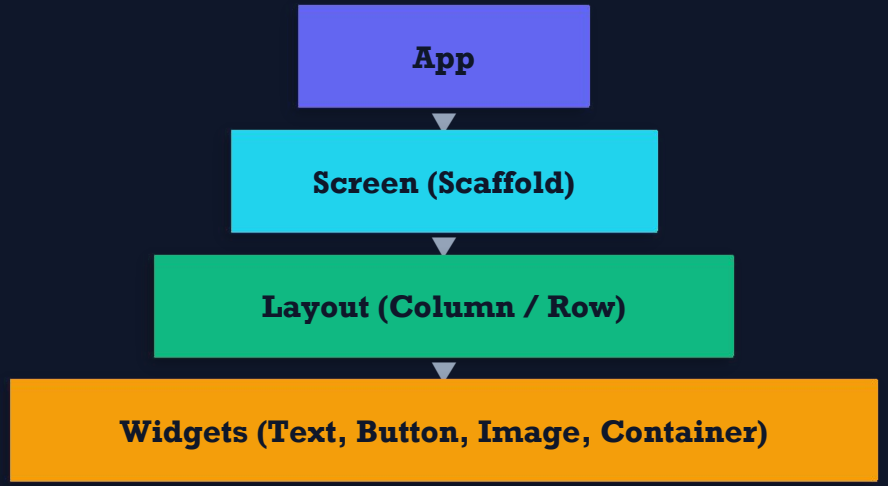
Single Codebase

Write once, deploy to Android, iOS, Web, and Desktop simultaneously.

Flutter Key Concept: Widgets

> *"Everything in Flutter is a Widget"*

Widget Hierarchy:



Common Widgets:

- Text
- Button
- Image
- Container
- Column
- Row
- ListView
- Icon

Who Uses Flutter?

Major global companies trust Flutter to build their production apps:

Google

Uses Flutter for internal tools and Google Pay

Alibaba

Xianyu app — 50M+ users built with Flutter

BMW

My BMW app for connected car experience

eBay

eBay Motors — Flutter-powered marketplace

Tencent

Multiple apps including NOW Live platform

Reflectly

Award-winning journaling app

🚀 **Flutter is used in 1M+ apps on Google Play and App Store worldwide**

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Installing Flutter SDK & Development Environment Setup

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Flutter Setup Roadmap

1

⬇ Download Flutter SDK

Visit flutter.dev → Download for Windows/Mac/Linux → Extract to C:\flutter

2

Add to PATH

Environment Variables → System PATH → Add C:\flutter\bin

3

Install Android Studio

Download Android Studio → Install Android SDK + Emulator → Add Flutter & Dart plugins

4

Run flutter doctor

Open terminal → Run: flutter doctor → Verify ✓ Flutter SDK ✓ Android ✓ VS Code

5

Create First App

flutter create my_first_app → cd my_first_app → flutter run

Understanding: flutter doctor

Terminal

```
$ flutter doctor

Doctor summary (to see all details, run flutter doctor -v):

[✓] Flutter (Channel stable, 3.x.x)

[✓] Android toolchain - develop for Android devices

[✓] Chrome - develop for the web

[✓] VS Code (version x.x.x)

• No issues found!
```

What each check means:

✓ Flutter SDK — Core framework installed and working

✓ Chrome — Web development capability enabled

✓ Android toolchain — Android SDK + ADB + build tools ready

✓ VS Code — Code editor with Flutter extension detected

Creating Your First Flutter App

```
$ flutter create my_first_app

  Creating project my_first_app...

$ cd my_first_app

$ flutter run
```

Project Structure

```
my_first_app/
  lib/
    main.dart ★
  android/
  ios/
  pubspec.yaml
```

Key File Concepts:

main.dart
Entry point of the app. Contains main() function and Widget tree.

MaterialApp
Top-level widget. Provides Material Design theme, navigation.

Scaffold
Basic screen structure: AppBar, Body, FloatingActionButton.

Hot Reload
Press 'r' in terminal → changes appear instantly without restart!

Week 1 — Complete Summary

Session 1 Overview & Platforms	Session 2 Flutter Introduction	Session 3 SDK & Environment Setup
✓ Mobile apps run on smartphones & tablets	✓ Native = best perf, one platform	✓ Download SDK → Set PATH → Install Android Studio
✓ Android (Google) vs iOS (Apple)	✓ Cross-platform = one codebase, many platforms	✓ flutter doctor verifies your setup
✓ Mobile ecosystem: OS → Store → Dev → User	✓ Flutter is Google's UI toolkit using Dart	✓ flutter create → flutter run = first app!
✓ App types: Native, Web, Cross-Platform	✓ Hot Reload + Widgets are Flutter's superpowers	✓ main.dart → MaterialApp → Scaffold → Widgets